



JD CORTEX

Perception, Reasoning, and Memory for an EZ-Robot JD Humanoid

Built on Synthiam ARC, Google Gemini, and YOLO vision

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Vision pipeline · Gemini reasoning · Whitelisted actions · Surveillance & ambient modes

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Witness memory · ARC control panel · Gesture control · System Launcher

What JD Cortex Actually Does



Understands you

Typed, spoken, or gestured commands are understood in plain language.



Watches the room

A camera pipeline notices who is present, their posture, and what they hold.



Remembers

Arrivals, departures, and sightings are kept in a running diary.

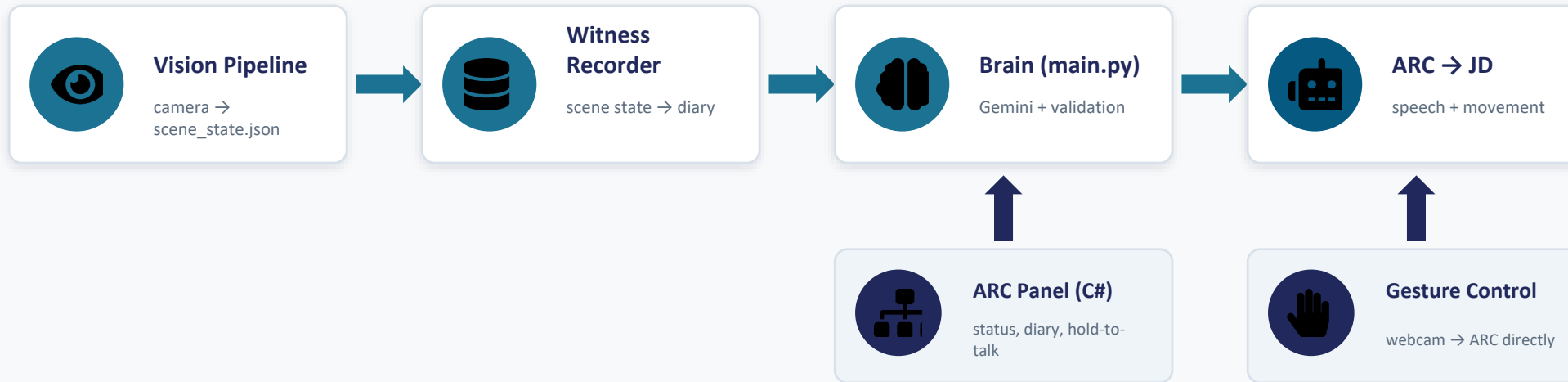


Stays safe

Every action is checked against a fixed, pre-approved whitelist before it runs.

The core idea: sensing, remembering, deciding, and acting are handled by separate processes. Nothing is done by one giant program — which makes the whole system easier to reason about, and safer to run.

Five Cooperating Processes



Why split it up? Only main.py ever opens the ARC connection that speaks — watchers queue messages through a shared file instead of talking to ARC directly. Gesture control gets its own, completely separate ARC connection, so it can never collide with conversation.

How One Command Reaches JD

1

Input arrives

Typed, spoken (local Parakeet), or via the ARC panel — all reach the same `process_command()`.

2

Local match first

The text is checked against the safe-action list before anything else runs.

3

Gemini, if needed

Scene summary, witness diary, and chat history are attached to the prompt.

4

Reply is spoken

`main.py` sends the reply to ARC's SayWait — the only process that ever does.

5

Action re-checked

Any suggested action is validated again against `known_actions.py` before ARC ever sees it.

6

JD moves

A confirmed movement runs, then JD returns to Standing automatically.

The Brain — `jd_robot_system/main.py`



main.py

single ARC
connection



One connection, one owner

`arc_connection.py` opens a single TCP socket to ARC — opened once, used for every command.



Three input paths, one brain

Console typing, local voice (Parakeet), and the ARC panel all funnel through `process_command()`.



Validate, then act

`known_actions.py` holds the whitelist — movements, sounds, lights. Nothing runs without a match.



Background alert thread

Polls a file-backed speech queue so surveillance and ambient watchers can still speak safely.

Gemini Integration — Three Distinct Uses



`understand()`

Conversation & action match

Returns REPLY + MATCH. Only ever suggests an action — main.py must still validate it before it runs.



`assess_alert()`

Surveillance judgement

Fails SAFE: if Gemini is unreachable, it alerts anyway with a generic line rather than staying silent.



`ambient_comment()`

Companion remarks

Fails SILENT: if Gemini is unreachable, it simply says nothing — a skipped remark costs nothing.

Quota resilience: each call walks a list of model names — skipping to the next on a 404 — and rotates through up to three API keys per model, since the free tier allows roughly 20 requests/day/key/model. A live usage counter is shown on the ARC panel.

Vision Pipeline — A Pure Sensor Process



Object detection

YOLO identifies general objects in frame.



Pose tracking

YOLO pose gives each person a tracker ID.



Face recognition

Matches against known_encodings.pkl to assign real names.

`vision_pipeline/scene_state.json`

```
{
  "timestamp": 1723222110.4,
  "objects_visible": ["laptop", "cup"],
  "people": {
    "3": {
      "name": "aima",
      "posture": "sitting",
      "holding": "cup",
      "sitting_on": "chair"
    }
  }
}
```

Written atomically: a temporary .tmp file is renamed into place with `os.replace(...)`, so no reader — scene_context, the witness recorder, or the panel — ever sees a half-written file.

The Witness Diary — A Robot That Remembers



SQLite
(WAL mode)

Identity by name, never tracker ID

The same person can burn through dozens of tracker IDs a minute — unrecognised sightings pool into one “unknown” bucket instead.

Debounced arrivals & departures

A couple of seconds of continuous presence confirms an arrival; a longer solid gap confirms a departure — no flapping on a tracking hiccup.

Feeds the same Gemini prompt

memory_context.py adds a diary block alongside the live scene — so “who did you see today?” is answered by the same brain as any other question.

Retention: 14 days by default (JD_DIARY_KEEP_DAYS) — pruned automatically at every recorder startup.

Hand-Gesture Control

Gesture → Action

Fist

Forward

Open hand

Stop

Index left / right

Turn

Index down

Sit

Peace sign

Wave

Three fingers

Stand

Index + pinky

Pushups



Reads a stationary webcam — never JD's own camera

JD's own camera moves when JD moves — a FORWARD gesture would shake the view, lose the hand, and trip the very safety stop meant to protect it.



Automatic safety stop

If the hand disappears mid-walk, JD is stopped automatically after a short number of hand-less frames — it can never be left walking indefinitely.

The ARC Control Panel (“Memory” Skill)

A C# ARC Behavior Control skill giving one live window into everything JD Cortex is doing.



Connection light

Live status of the link to the Python brain



Scene card

What JD currently sees, updated live



One input box

Free text for both memory questions and commands



Hold-to-talk

Push-to-record voice input from the panel



Gesture toggle

Starts/stops hand-gesture control with a plain-language legend



Guarded shutdown

A confirmed stop button for the whole system

Protocol: 127.0.0.1:5005 — one line in, one line out (status, diary [n], alerts [n], gesture on/off, listen start/stop, stop, or free text).

Two Watchers, Two Failure Philosophies



Surveillance Watcher

Fails **SAFE**

- Watches for unknown people and sensitive objects (knives, scissors)
- A condition must persist across multiple polls before it fires
- Logs to activity_log.txt and requests a snapshot
- If Gemini is unreachable, it alerts anyway with generic wording — silence could hide a real event



Ambient Watcher

Fails **SILENT**

- Watches for meaningful changes — arrivals, departures, posture, idle time
- Asks Gemini whether a short remark is genuinely natural right now
- Deliberately conservative — silence is the common, expected case
- If Gemini is unreachable, it just says nothing — a skipped remark costs nothing

Tech Stack & Running the System

Vision

YOLO (ultralytics) · OpenCV · face_recognition

Reasoning

Google Gemini — plain HTTP, no SDK

Speech

Local Parakeet (sherpa-onnx) — nothing leaves the machine

Gestures

MediaPipe 0.10.21 (pinned)

Memory

SQLite, WAL mode

Control

Synthiam ARC — TCP script server, port 6666

One launcher for the whole system

```
python run.py check
```

reports every library, model file, key, and port before a session

```
python run.py
```

starts vision, memory, brain, and watchers in dependency order

```
python run.py --minimal
```

brain + witness memory only, no vision

Thank You

JD Cortex keeps sensing, remembering, deciding, and acting as separate, cooperating processes — safe by construction, not by afterthought.

- One whitelist, always checked — Gemini reasons, but never drives JD directly.
- One ARC voice, one owner — every other process queues speech instead of speaking itself.
- Memory by name, not tracker ID — durable, readable, and honest about what JD actually saw.

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Project: JD Cortex